Mixed frequency, MIDAS and dynamic factors

Status: draft for review

Intuition

Monthly indicators and quarterly GDP do not share a calendar. A nowcast asks what the available monthly releases imply for the still-unpublished quarter. Missing recent observations form a ragged edge. The correct procedure models this information set; interpolating unpublished observations pretends information exists.

Model

A bridge uses $g_q = a + b' \bar{x}_q + e_q$. U-MIDAS uses separate coefficients for month slots. A factor model writes $x_t = \Lambda f_t + \epsilon_t$, $f_t = A f_{t-1} + v_t$ with quarterly aggregation restrictions linking GDP to monthly latent growth.

Assumptions and interpretation

A bridge aggregates monthly information, while an unrestricted MIDAS design retains within-quarter timing. Both need a rule for unavailable slots. State-space filtering and smoothing condition on released observations and can accommodate missing entries. Smoothing is legitimate for a nowcast if the entire supplied observation set is bounded by the origin; smoothing with later released information leaks the future.

Application and limits

Fixed release delays are a transparent approximation, not a reconstructed calendar. Training GDP targets must also satisfy release timing. Convergence must be checked rather than inferred from a returned result object. One factor can miss sector-specific information; more factors increase variance and computational burden. COVID-sensitive GDP errors should be reported with a declared exclusion comparison, without choosing a preferred period after looking at scores.

Exam questions and answers

1. What is a ragged edge?
Answer. Different indicators end at different dates because releases have different delays or gaps.

2. How does U-MIDAS differ from a bridge?
Answer. It estimates separate month-slot effects rather than a single within-quarter average.

3. When does smoothing leak?
Answer. When observations released after the simulated origin are included.

4. Why hide GDP after quarter end?
Answer. The quarter may have ended before its GDP estimate is published.

5. Does a finite forecast prove convergence?
Answer. No; likelihood and iteration diagnostics must be inspected.

Reasoning exercise

A quarter has ended, but its GDP estimate is unpublished and the most recent industrial-production month is missing. Is filling that month by linear interpolation compatible with the information set? Answer: a future endpoint is unavailable, and a filled value would conceal genuine uncertainty. Keep the source gap, use a declared training-only design rule for bridge/MIDAS, and let the state-space model condition on the actually released series. An origin-bounded smoother may use all releases already available; it must not ingest later observations. Check numerical convergence separately from whether a finite nowcast was returned. Compare methods on common quarters and retain a declared pandemic sensitivity.

Verified reference

https://doi.org/10.1002/jae.2306